

10.2 DC: Potential Dividers

Question Paper

Course	CIE A Level Physics
Section	10. D.C. Circuits
Topic	10.2 DC: Potential Dividers
Difficulty	Easy

Time allowed: 40

Score: /27

Percentage: /100

Question 1a

Sketch the circuits symbols for

(i)
the potentiometer

[1]

(ii)
the galvanometer

[1]

[2 marks]

Question 1b

Fig. 1.1 shows a circuit with a potentiometer.

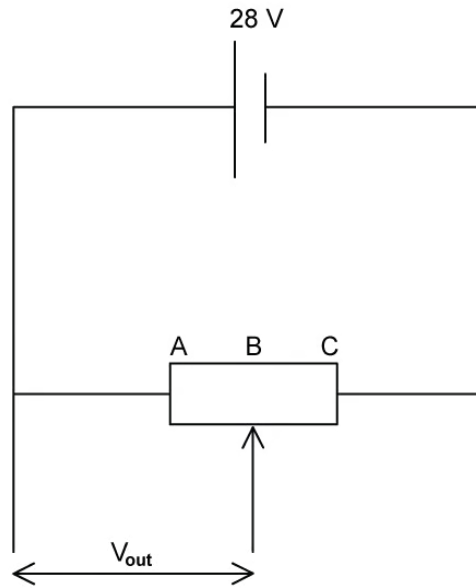


Fig. 1.1

Fixed ends AC are connected across the battery so that there is a full battery voltage across the whole resistor.

State the voltage output, V_{out} when the sliding connect is at

(i)

A

[1]

(ii)

B

[1]

(iii)

C

[1]

[3 marks]

Question 1c

The resistor is now replaced with a uniform resistance wire connected between ends AC, as shown in Fig. 1.2a. The 28 V cell is replaced with a cell of unknown e.m.f.

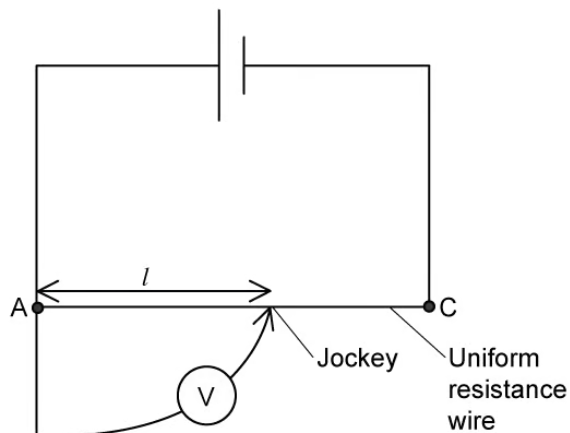


Fig. 1.2a

Sketch a graph of Fig 1.2b to show the voltmeter reading V against length l as the jockey is moved from point A to point C.

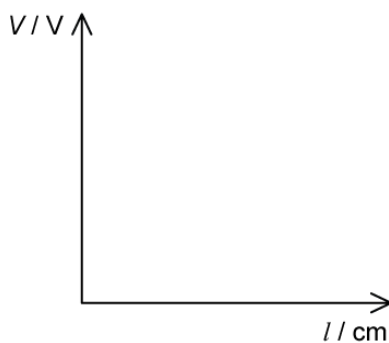


Fig 1.2b

[2 marks]

Question 1d

A potentiometer can be used to compare e.m.f.s of cells or potential differences.

A test cell is added to the circuit in Fig. 1.2a. A student sets up the circuit shown in Fig. 1.3.

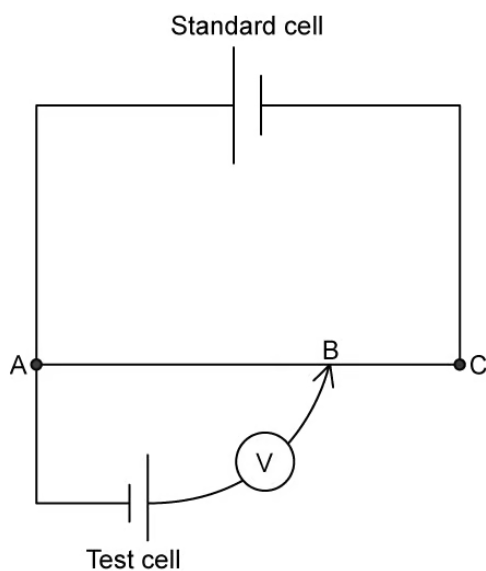


Fig. 1.3

(i)

State why the student will not be able to find a point where the current is zero.

[1]

(ii)

The student fixes the issue from **(a)(i)** and achieves a balance point when the length of the wire AB is 25.6 cm.

They repeat the experiment with a standard cell of e.m.f 1.545 V. The balance point using this cell is 37.2 cm.

Calculate the e.m.f of the test cell.

[3]

[4 marks]

Question 2a

Fig 1.1 shows a potential divider circuit containing a fixed resistor and thermistor.

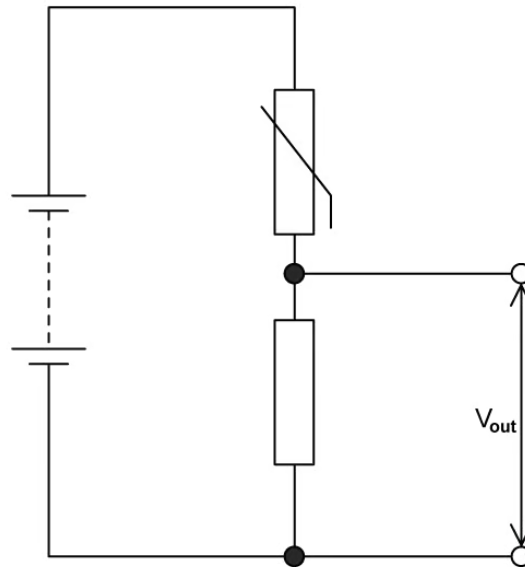


Fig 1.1

Explains what happens as the temperature increases.

[2 marks]

Question 2b

The thermistor is now replaced with an LDR and the output voltage is now taken from the LDR, as shown in Fig. 1.2a.

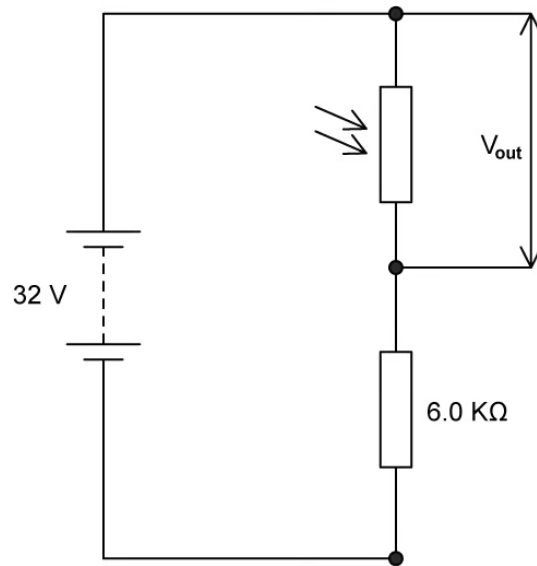


Fig. 1.2a

Using Fig. 1.2b, determine the resistance of the LDR when the light intensity is 120 lux.

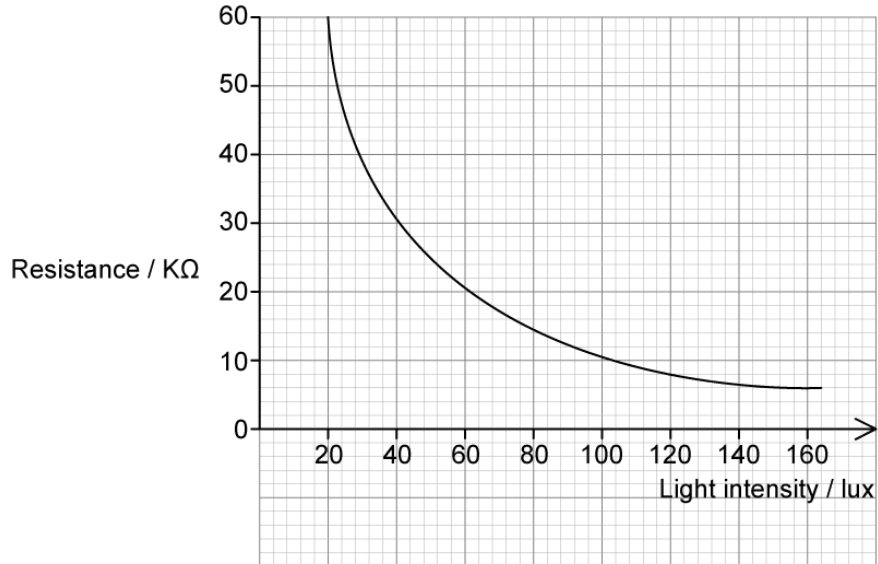


Fig. 1.2b

[1 mark]

Question 2c

Using your answer to part (b), calculate V_{out} when the light intensity is 120 lux.

[3 marks]**Question 2d**

Using your answer to part (c), determine the voltage across the $6.0 \text{ k}\Omega$ resistor at a light intensity of 120 lux.

[2 marks]

Question 3a

A variable resistor is used to control the circuit shown in Fig. 1.1.

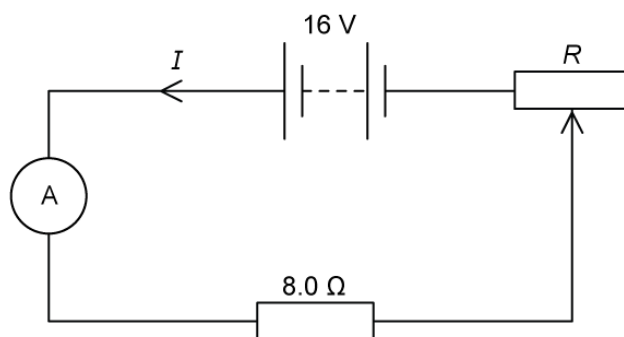


Fig. 1.1

The variable resistor is connected in series with a 16 V power supply of negligible internal resistance, an ammeter and an 8.0 Ω resistor.

The resistance R of the variable resistor can be varied between 0 to 12 Ω .

Calculate

(i)
the minimum possible current

[2]

(ii)
the maximum possible current

[2]

[4 marks]

Question 3b

On Fig 1.2, sketch the variation with R of current I in the circuit.

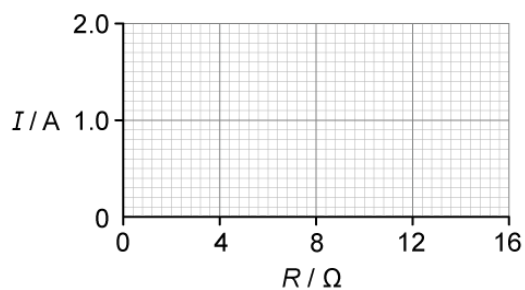


Fig. 1.2

[2 marks]

Question 3c

Galvanometers are used in *null methods*.

Explain how a galvanometer is used in *null methods*.

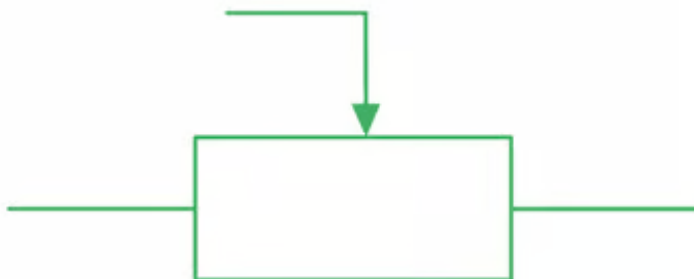
[2 marks]

Model Answers: Easy

1a

The circuit symbols for:

(i) the potentiometer is:



- Correct circuit symbol as shown above; [1 mark]

(ii) the galvanometer is:



- Correct circuit symbol as shown above; [1 mark]

[Total: 2 marks]

1b

The voltage output at:

(i) A is:

- 0 V [1 mark]

(ii) B is:

- 14 V [1 mark]

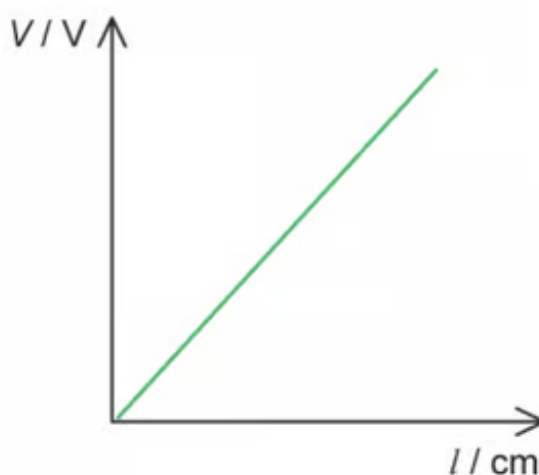
(iii) C is:

- 28 V [1 mark]

[Total: 3 marks]

1c

A graph of the voltmeter reading V and length l of the jockey as it is moved from A to C is



- Straight line with a positive gradient; [1 mark]
- Through the origin; [1 mark]

[Total: 2 marks]

1d

(i) The student will not be able to find a point where the current is zero because:

- The test cell is the wrong way around / the test cell needs to be reversed; [1 mark]

(ii) The e.m.f of the test cell is:

List the known quantities:

- Length AB at first balance point, $l_1 = 25.6$ cm
- E.m.f of standard cell, $E_2 = 1.545$ V
- Length AB at second balance point, $l_2 = 37.2$ cm

Determine the equation linking the length and e.m.f.s:

- $\frac{l_1}{l_2} = \frac{\text{E.m.f of test cell}}{E_2}$ [1 mark]
- $\text{E.m.f of test cell} = E_2 \times \frac{l_1}{l_2}$

Substitute in the values:

- $\text{E.m.f of test cell} = 1.545 \times \frac{25.6}{37.2}$ [1 mark]
- $\text{E.m.f of test cell} = 1.06$ V [1 mark]

[Total: 3 marks]

2a

As the temperature increases:

- The resistance of the thermistor decreases; [1 mark]
- So V_{out} increases; [1 mark]

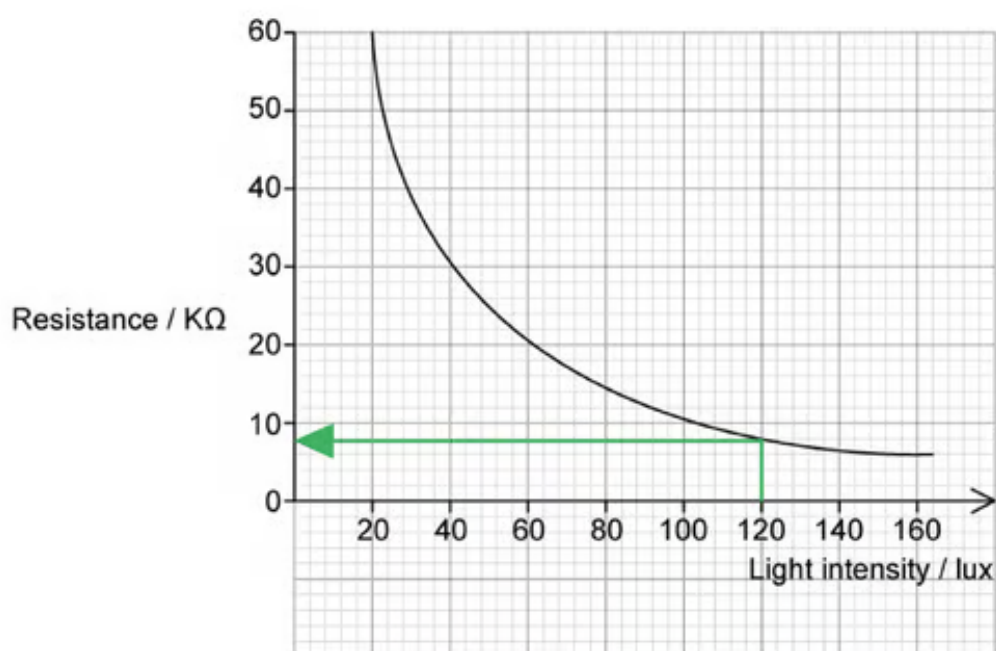
[Total: 2 marks]

Potential dividers are all about the **sharing** of the potential from the battery (hence the name). When the resistance of the thermistor decreases, this means the voltage through it also decreases (from $V = IR$ where I is the same for both resistors).

However, the battery has a set voltage. Therefore, the voltage of the fixed resistor must **increase** so the combined voltage of the thermistor and fixed resistor remains the same. V_{out} is the reading of the voltage from the fixed resistor.

2b

The resistance of the LDR when the light intensity is 120 lux is:



- 8 kΩ [1 mark]

[Total: 1 mark]

2c

V_{out} when the light intensity is 120 lux is:

List the known quantities:

- Resistance of the LDR, $R_1 = 8.0 \text{ k}\Omega$ (from part b)
- Resistance of the fixed resistor, $R_2 = 6.0 \text{ k}\Omega$

- Voltage of the power supply, $V_{in} = 32 \text{ V}$

Determine the equation for V_{out} :

- $V_{out} = \frac{R_1}{R_1 + R_2} \times V_{in}$ [1 mark]

Substitute in the values:

- $V_{out} = \frac{8.0}{8.0 + 6.0} \times 32$ [1 mark]
- $V_{out} = 18.2857 = 18 \text{ V}$ [1 mark]

[Total: 3 marks]

2d

The voltage across the $6.0 \text{ k}\Omega$ resistor is:

List the known quantities:

- Voltage across the LDR, $V = 18 \text{ V}$

List the known quantities:

- Resistance of the LDR, $R_1 = 8.0 \text{ k}\Omega$ (from part b)
- Resistance of the fixed resistor, $R_2 = 6.0 \text{ k}\Omega$
- Voltage of the power supply, $V_{in} = 32 \text{ V}$

Determine the equation for V_{out} :

- $V_{out} = \frac{R_1}{R_1 + R_2} \times V_{in}$ [1 mark]

Substitute in the values:

- $V_{out} = \frac{8.0}{8.0 + 6.0} \times 32$ [1 mark]
- $V_{out} = 18.2857 = 18 \text{ V}$ [1 mark]

[Total: 3 marks]

2d

The voltage across the $6.0 \text{ k}\Omega$ resistor is:

List the known quantities:

- Voltage across the LDR, $V_{out} = 18 \text{ V}$
- Voltage of the power supply, $V_{in} = 32 \text{ V}$

Calculate the voltage across the $6.0 \text{ k}\Omega$ resistor:

- $V_{6.0\text{k}\Omega} = V_{in} - V_{out} = 32 - 18$ [1 mark]
- $V_{6.0\text{k}\Omega} = 14 \text{ V}$ [1 mark]

[Total: 2 marks]

3a

(i) Calculate the minimum possible current:

- $V = IR \Rightarrow I = \frac{V}{R}$

- $I \propto \frac{1}{R}$ so minimum current occurs with maximum resistance
- In series: $R_{\max} = 8 + 12 = 20 \, \Omega$ [1 mark]

Calculate the minimum current:

- $I_{\min} = \frac{16}{20}$ [1 mark]
- $I_{\min} = 0.8 \, \text{A}$ [1 mark]

(ii) Calculate the maximum possible current:

- Maximum current occurs with minimum resistance, $R_{\min} = 8 \, \Omega$
- $I_{\max} = \frac{16}{8}$ [1 mark]
- $I_{\max} = 2.0 \, \text{A}$ [1 mark]

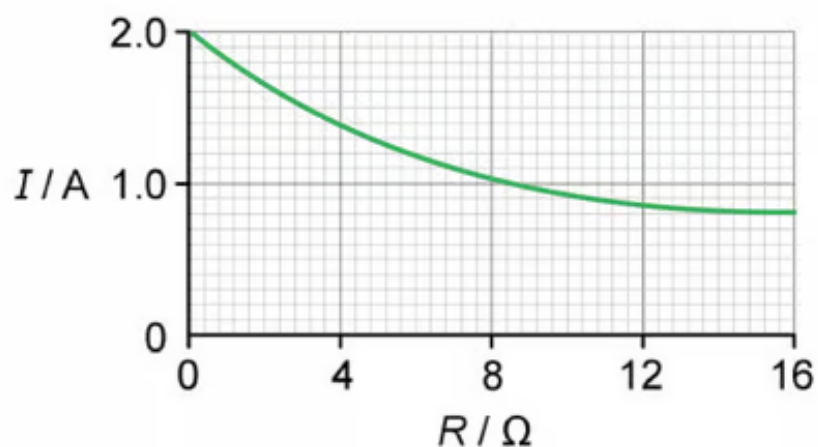
[Total: 4 marks]

3b

The variation with R of current I in the circuit should look like:

List the known quantities:

- Maximum current = 2.0 A (from part a)
- Minimum current = 0.8 A (from part a)

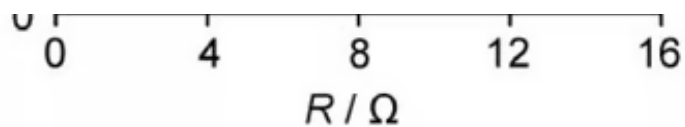


- Starting point at (0, 2.0) and end point at (16, 0.8); [1 mark]
- Curve with decreasing gradient; [1 mark]

[Total: 2 marks]

3c

Galvanometers are used in null methods by:



- Starting point at (0, 2.0) and end point at (16, 0.8); [1 mark]
- Curve with decreasing gradient; [1 mark]

[Total: 2 marks]

3c

Galvanometers are used in null methods by:

- The galvanometer has no deflection; [1 mark]
- When there is no current flow / no potential difference; [1 mark]

[Total: 2 marks]

